(Assignment 1) GEN AI

Question (3)

Forward Pass:-

$X = [[0.34], [0.55]]$

$W^1 = [[0.42, 0.72], [0, 0.3], [0.15, 0.09]]$

$W^2 = [[0.44, 0.72], 0.44, 0.03, 0.55]]$

$W^3 = [[0.55], [0.71], [0.29]]$

$W^4 = [[0.99, 0.55, 0.99], [0.71, 0.7, 0.22]]$

L.R = 0.5

$$O^1 = w^1 \times \begin{bmatrix} 0.42 & 0.72 \\ 0 & 0.3 \\ 0.15 & 0.09 \end{bmatrix} \begin{bmatrix} 0.34 \\ 0.55 \end{bmatrix}$$

$$O^1 = \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix} \Rightarrow Relu(O^1)$$

$$= [0.44 \quad 0.03 \quad 0.55]$$

$O^2 =$ ~~0.5388~~ ~~0.165~~ ~~0.1005~~ $= \{0.44 \quad 0.03 \quad 0.55\}$

$\times \begin{bmatrix} 0.5388 \\ 0.165 \\ 0.1005 \end{bmatrix}$

$O^2 = [0.2972]$

$O^3 =$

$O^3 = \begin{bmatrix} 0.55 \\ 0.71 \\ 0.29 \end{bmatrix} \cdot [0.2972] = \begin{bmatrix} 0.1634 \\ 0.2116 \\ 0.0861 \end{bmatrix}$

$O^4 = \begin{bmatrix} 0.97 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix} \begin{bmatrix} 0.1634 \\ 0.2116 \\ 0.0361 \end{bmatrix}$

$O^4 = \begin{bmatrix} 0.3586 \\ 0.2826 \end{bmatrix}$

Reconstruction cost :-

$$J(\theta) = \sum_{i=1}^{x} x_i \log(\hat{x}_i)$$

$$= -0.34 \log_e (0.3580) - 0.55 (\log_e 0.2820)$$

$$= 1.414$$

Backward Pass :-

$$\Delta O^4_{DIV} = O^4 - \text{output}$$

$$\begin{bmatrix} 0.3580 \\ 0.2820 \end{bmatrix} - \begin{bmatrix} 0.34 \\ 0.55 \end{bmatrix} = \begin{bmatrix} 0.018 \\ -0.2674 \end{bmatrix}$$

$$dw = \begin{bmatrix} 0.018 \\ -0.2774 \end{bmatrix} \begin{bmatrix} 0.1639 & 0.210 & 0.0861 \end{bmatrix}$$

$$= \begin{bmatrix} 0.002 & 0.0003 & 0.001 \\ -0.043 & -0.056 & -0.023 \end{bmatrix}$$

$$\Delta \nabla O^3_{DIV} = \begin{bmatrix} -0.174 \\ -0.178 \\ -0.648 \end{bmatrix}$$

$$dw_3 = \begin{bmatrix} -0.051 \\ -0.52 \\ -0.012 \end{bmatrix}$$

$A \nabla O^2 Div = [-0.243]$

$dw_{22} = [-0.126 \quad -0.038 \quad -0.028]$

$\nabla O' Div = \begin{bmatrix} -0.102 \\ -0.007 \\ -0.128 \end{bmatrix}$

$dw_{11} = \begin{bmatrix} -0.034 & -056 \\ 0.002 & 0.003 \\ 0.043 & 0.07 \end{bmatrix}$

New weights

$w_{21} = \begin{bmatrix} 0.469 & 0.548 & 0.469 \\ 0.731 & 0.728 & 0.231 \end{bmatrix}$

$w_{32} \begin{bmatrix} 0.575 \\ 0.736 \\ 0.796 \end{bmatrix}$

$w_{22} [0.503 \quad 0.049 \quad 0.561]$

$w_1 \begin{bmatrix} 0.437 & 0.748 \\ 0.001 & 0.301 \\ 0.171 & 0.125 \end{bmatrix}$

Question (03) :-

Forward Pass

$$Z_1 = V_1 = \begin{bmatrix} 0.5386 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$u' = \begin{bmatrix} 0.5 & 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5386 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$u' = [0.402] = u$$

$$b' = \begin{bmatrix} 0.5 & 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0.5386 \\ 0.165 \\ 0.1005 \end{bmatrix}$$

$$b' = [0.402] = b$$

$$z = u + b \cdot \varepsilon \qquad : \varepsilon = 1 \text{ (Assume)}$$

$$Z_2 = [0.402] + [0.402] = 0.804$$

$$V_3 = Z_3 = \begin{bmatrix} 0.5 & 5 \\ 0.71 \\ 0.29 \end{bmatrix} [0.804] = \begin{bmatrix} 0.442 \\ 0.57 \\ 0.233 \end{bmatrix}$$

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$$Z_4 = \begin{bmatrix} 0.97 & 0.55 & 0.97 \\ 0.71 & 0.7 & 0.22 \end{bmatrix} \begin{bmatrix} 0.442 \\ 0.57 \\ 0.233 \end{bmatrix}$$

$$Y_4 = Z_4 = \begin{bmatrix} 0.968 \\ 0.764 \end{bmatrix}$$

$$L = -0.34 \log(0.968) - 0.55 \log(0.764)$$
$$L = 0.159$$

$$|cL| = -1/2 (1 + \log \sigma^2 - \mu^2 - \sigma^2)$$
$$|cL| = -1/2 (1 + \log(0.6\sigma^2) - 0.4\sigma^2 - 0.4\sigma^2)$$
$$|cL| = 0.572$$

$$J(\theta) = L + |cL| = 0.159 + 0.572 = 0.731$$

In back-ood pass $\Delta Y_4 Div$, $\Delta Z_4 Dv$, dw_4, $\Delta y^3 Dv$, $\Delta Z_3 Dv$ will be calculated just like autoencoder. From dw_4, all of them will be calculated Dos

$$dw_4 = \Delta Z_4 Div \times Z_3^T$$
$$\nabla Z_3 Div = W_4^T \times \Delta Z_4 Dv$$
$$\nabla U Dv = 1 \times \nabla Z_3 Dv \cdot U$$
$$\nabla U' Dv = \nabla U U \times \nabla U Dv$$
$$\nabla b Dv = \varepsilon \times \nabla Z_3 Div \cdot (b - 1/b)$$

$\nabla 6' Dir = [1] \times \nabla 6 Dir$

$\partial w_{11} = \nabla 11' Dir \times y^T$

$\partial w_0 = A \nabla 6' Dir \times x^T$

$\nabla y^T Dir = w_{11}^T \times \nabla 11' Dir + w_0^T \times A \nabla 6' Dir$

$\nabla 2_1 Dir = \nabla 2_1 y' \times \nabla y_1 Dir$

$\partial w_1 = \nabla 2_1 Dir \times x^T$

$\nabla 2_1 Dir$ and ∂w_1 will be calculated similarly as we done in autoencoder hence, we can update weights by using this method.

Thank you